Shivling Bhange

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Profile

Frontend Developer with **5+ years of experience** building **high-performance, scalable** web applications for enterprise solutions. Proven track record in reducing report generation time by 60-65% for large datasets and improving application load times by 2x through React performance optimization and efficient data parsing. Expert in **React.js**, **TypeScript**, and AWS, known for mentoring junior engineers and delivering robust, maintainable software through **TDD/BDD** practices and clean component architecture.

Skills

Frontend: HTML5, **React.js, Redux**, React Hooks, Context API, Angular, JavaScript (ES6+), **TypeScript**, CSS3, MUI, Bootstrap, React Router, Axios, Webpack, Next.js, **Redux Toolkit** . Micro Frontend,

Database: MongoDB .

DevOps & Tools: Git, GitHub, Bitbucket, Jira, Git Action, AWS (EC2, S3, Lambda), Docker Basics, Postman .

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Testing: Jest, React Testing Library, Mocha, Chai, **BDD** (Gherkin), **TDD**, Unit Testing, Integration Testing .

Other: GitHub Copilot/AI-assisted development, CI/CD, Agile/Scrum .

Professional Experience

Software Engineer - BMW Techworks India **(BMW Group)** *01/2024 – Present | Pune, India*

- Improved system performance by reducing report generation time by 60-65% for 150-200MB datasets through **memoization** and pagination techniques, enabling faster data processing for engineering teams.
- Designed and implemented a diff report to detect added, removed, and modified ECU parameters by comparing two DCM files, reducing manual analysis of 200K+ parameters from 2-3 hours to under 5 seconds.
- Understanding of accessibility principles, Section **508 compliance**, and **WCAG** guidelines .
- Developed a high-precision ECU comparison tool using data structure & algorithms to optimize rendering pipelines, applying analytical precision to debugging complex parameter changes for 200+ engineers, reducing analysis time for **200K+ parameters** to under **5** seconds.
- Led code reviews, mentored junior developers, and established testing best practices using Jest, React Testing Library, Mocha, and Chai while leveraging AI-assisted development tools such as GitHub Copilot and LLM-based workflows to improve development productivity and code quality.
- Managed complex application state with Redux Toolkit and Context API, implementing typed slices, selectors, and
- **Mentored** and guided junior engineers through **code reviews, pair programming**, and technical knowledge-sharing sessions, fostering a culture of continuous learning and improving overall code quality across the team.

- middleware to ensure predictable state flow and improve performance across data-heavy UI components.
- Implemented Role-Based Access Control (**RBAC**) with Admin and Application Developer roles for 200+ users, ensuring secure access and user-level permissions in compliance with ASPICE and ISO standards.
- Built advanced UI tools including search, filters, and visual signal comparisons using React.js and WASM, improving workflow efficiency by 70% for 200+ engineers.
- Built a reusable React component library with 500+ unit and integration tests using Jest, React Testing Library, and Mocha following TDD, BDD practices, achieving 90% code coverage and significantly reducing regression bugs across the application.

Software Engineer - Colortokens Inc *06/2022 – 01/2024 | Banglore, IND*

- Designed a centralized admin dashboard to manage and monitor 100K+ endpoints across the network, displaying hostname, IP, groups, firewall rules, policies, and system status, enabling real-time visibility and reducing manual monitoring effort.
- Integrated with X-Access VPN desktop application to enable secure token-based authentication, access control, and endpoint-level policy enforcement, streamlining onboarding for the entire user base while strengthening security through authenticator-based verification.
- Optimized REST API performance through indexing and query optimization using NodeJS, reducing initial load time for 200K+ records from 6 seconds to 3 seconds, achieving 2x faster data retrieval.
- Implemented remote resource decommissioning to securely offboard endpoints, automatically deleting system policies, access tokens, and associated data in the background, ensuring proper offboarding and compliance.
- Developed audit logs, alerts, role-based permissions, and alert history for security compliance, reducing audit log load time by 70% through API optimization.

Software Engineer – GravityTech Software Solution *01/2021 – 06/2022 | Pune, IND*

- Engineered secure and scalable internal web applications for a **multinational banking** institution, ensuring compliance with industry-grade security protocols and data protection standards.
- Developed global access portals serving internal clients across multiple regions, enabling seamless cross-location collaboration and centralized access to critical banking resources.
- Delivered high-performance applications while ensuring strict adherence to internal security protocols and banking industry compliance standards.
- Implemented Multi-Factor Authentication **(MFA**) to enhance security during user registration and account creation in web and mobile applications, reducing unauthorized access risks.

Education

SGGS Institute of Engineering & Technology (Gov. Autonomous) *06/2019*

B.Tech. (Electronics & Telecommunication)

Certificates

Test-Driven Development for JavaScript

Introduction to Agile Development and Scrum